

Отчет по лабораторной номер 5

1) Для заданной на схеме schema-lab5 сети, состоящей из управляемых коммутаторов, маршрутизаторов и персональных компьютеров

выполнить планирование и документирование адресного пространства и назначить статические адреса всем устройствам.

nb! Каждое соединение маршрутизатора с маршрутизатором - это отдельная сеть.

R4:

enable

configure terminal

hostname R4

Интерфейс к PC1

interface FastEthernet0/0

ip address 192.168.1.1 255.255.255.0

no shutdown

exit

Интерфейс к R5

interface FastEthernet1/0

ip address 10.0.45.1 255.255.255.252

no shutdown

exit

end

write memory

R5:

enable

configure terminal

hostname R5

Интерфейс к PC2

```
interface FastEthernet0/0
ip address 192.168.2.1 255.255.255.0
no shutdown
exit
```

Интерфейс к R4

```
interface FastEthernet1/0
ip address 10.0.45.2 255.255.255.252
no shutdown
exit
```

Интерфейс к R1

```
interface FastEthernet2/0
ip address 10.0.15.2 255.255.255.252
no shutdown
exit
end
write memory
```

R1:

```
enable
configure terminal
hostname R1
```

Интерфейс к R5

```
interface FastEthernet0/0
ip address 10.0.15.1 255.255.255.252
no shutdown
exit
```

Интерфейс к Switch

```
interface FastEthernet1/0
```

ip address 10.0.0.1 255.255.255.0

no shutdown

exit

end

write memory

R2:

enable

configure terminal

hostname R2

Интерфейс к Switch

interface FastEthernet0/0

ip address 10.0.0.2 255.255.255.0

no shutdown

exit

Интерфейс к R6

interface FastEthernet1/0

ip address 10.0.26.1 255.255.255.252

no shutdown

exit

end

write memory

R3:

enable

configure terminal

hostname R3

Интерфейс к Switch

interface FastEthernet0/0

```
ip address 10.0.0.3 255.255.255.0
```

```
no shutdown
```

```
exit
```

Интерфейс к R7

```
interface FastEthernet1/0
```

```
ip address 10.0.37.1 255.255.255.252
```

```
no shutdown
```

```
exit
```

Интерфейс к R8

```
interface FastEthernet2/0
```

```
ip address 10.0.38.1 255.255.255.252
```

```
no shutdown
```

```
exit
```

```
end
```

```
write memory
```

R8:

```
enable
```

```
configure terminal
```

```
hostname R8
```

Интерфейс к R3

```
interface FastEthernet0/0
```

```
ip address 10.0.38.2 255.255.255.252
```

```
no shutdown
```

```
exit
```

Интерфейс к PC5

```
interface FastEthernet1/0
```

```
ip address 192.168.5.1 255.255.255.0
```

no shutdown
exit
end
write memory

R6:

enable
configure terminal
hostname R6

Интерфейс к R2

interface FastEthernet0/0
ip address 10.0.26.2 255.255.255.252
no shutdown
exit

Интерфейс к PC3

interface FastEthernet1/0
ip address 192.168.3.1 255.255.255.0
no shutdown
exit

Интерфейс к R7

interface FastEthernet2/0
ip address 10.0.67.1 255.255.255.252
no shutdown
exit
end
write memory

R7:

enable

configure terminal

hostname R7

Интерфейс к R3

interface FastEthernet0/0

ip address 10.0.37.2 255.255.255.252

no shutdown

exit

Интерфейс к PC4

interface FastEthernet1/0

ip address 192.168.4.1 255.255.255.0

no shutdown

exit

Интерфейс к R6

interface FastEthernet2/0

ip address 10.0.67.2 255.255.255.252

no shutdown

exit

end

write memory

PC1> ip 192.168.1.10 255.255.255.0 192.168.1.1

```
PC1> ip 192.168.1.10 255.255.255.0 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1
```

PC2> ip 192.168.2.10 255.255.255.0 192.168.2.1

```
PC2> ip 192.168.2.10 255.255.255.0 192.168.2.1
Checking for duplicate address...
PC2 : 192.168.2.10 255.255.255.0 gateway 192.168.2.1
```

PC3> ip 192.168.3.10 255.255.255.0 192.168.3.1

```
PC3> ip 192.168.3.10 255.255.255.0 192.168.3.1
Checking for duplicate address...
PC3 : 192.168.3.10 255.255.255.0 gateway 192.168.3.1
```

PC4> ip 192.168.4.10 255.255.255.0 192.168.4.1

```
PC4> ip 192.168.4.10 255.255.255.0 192.168.4.1
Checking for duplicate address...
PC4 : 192.168.4.10 255.255.255.0 gateway 192.168.4.1
```

PC5> ip 192.168.5.10 255.255.255.0 192.168.5.1

```
PC5> ip 192.168.5.10 255.255.255.0 192.168.5.1
Checking for duplicate address...
PC5 : 192.168.5.10 255.255.255.0 gateway 192.168.5.1
```

2) Настроить протокол динамической маршрутизации RIP v2 для области, указанной на схеме schema-lab5.

R4:

enable

configure terminal

Включаем процесс RIP

router rip

Используем версию 2

version 2

Объявляем сеть PC1

network 192.168.1.0

Объявляем все сети 10.x.x.x

network 10.0.0.0

Отключаем автосуммаризацию

no auto-summary

exit

end

write memory

R5:

```
enable
configure terminal
router rip
version 2
network 192.168.2.0
network 10.0.0.0
no auto-summary
exit
end
write memory
```

R1:

```
enable
configure terminal
router rip
version 2
network 10.0.0.0
no auto-summary
exit
end
write memory
```

3) Настроить протокол динамической маршрутизации OSPF для зон 0, 1, 2.
Зону 1 настроить как полностью (nb!) тупиковую.

R1

```
enable
configure terminal
router ospf 1
network 10.0.0.0 0.0.0.255 area 0
```


exit

end

write memory

R2

enable

configure terminal

router ospf 1

network 10.0.0.0 0.0.0.255 area 0

network 10.0.26.0 0.0.0.3 area 2

exit

end

write memory

R3

enable

configure terminal

router ospf 1

network 10.0.0.0 0.0.0.255 area 0

network 10.0.38.0 0.0.0.3 area 1

network 10.0.37.0 0.0.0.3 area 2

area 1 stub no-summary

exit

end

write memory

R8

enable

configure terminal

router ospf 1

```
network 10.0.38.0 0.0.0.3 area 1
network 192.168.5.0 0.0.0.255 area 1
area 1 stub
exit
end
write memory
```

R6

```
enable
configure terminal
router ospf 1
network 10.0.26.0 0.0.0.3 area 2
network 10.0.67.0 0.0.0.3 area 2
network 192.168.3.0 0.0.0.255 area 2
exit
end
write memory
```

R7

```
enable
configure terminal
router ospf 1
network 10.0.37.0 0.0.0.3 area 2
network 10.0.67.0 0.0.0.3 area 2
network 192.168.4.0 0.0.0.255 area 2
exit
end
write memory
```

4) Настроить редистрибуцию маршрутов между протоколами RIP v2 и OSPF.

R1:

enable

configure terminal

Направление 1: Из RIP в OSPF

router ospf 1

redistribute rip metric 10 subnets

exit

Направление 2: Из OSPF в RIP

router rip

redistribute ospf 1 metric 1

exit

end

write memory

5) Проверить работоспособность маршрутизации, выполнив ping VPC "все между всеми" (nb!: в обе стороны).

PC1> ping 192.168.2.10

```
PC1> ping 192.168.2.10

192.168.2.10 icmp_seq=1 timeout
84 bytes from 192.168.2.10 icmp_seq=2 ttl=62 time=29.227 ms
84 bytes from 192.168.2.10 icmp_seq=3 ttl=62 time=25.678 ms
84 bytes from 192.168.2.10 icmp_seq=4 ttl=62 time=25.502 ms
84 bytes from 192.168.2.10 icmp_seq=5 ttl=62 time=25.847 ms
```

PC2> ping 192.168.1.10

```
PC2> ping 192.168.1.10

84 bytes from 192.168.1.10 icmp_seq=1 ttl=62 time=29.132 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=62 time=25.126 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=62 time=26.138 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=62 time=25.473 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=62 time=24.768 ms
```

PC1> ping 192.168.5.10

```
PC1> ping 192.168.5.10

192.168.5.10 icmp_seq=1 timeout
84 bytes from 192.168.5.10 icmp_seq=2 ttl=59 time=66.563 ms
84 bytes from 192.168.5.10 icmp_seq=3 ttl=59 time=65.287 ms
84 bytes from 192.168.5.10 icmp_seq=4 ttl=59 time=54.992 ms
84 bytes from 192.168.5.10 icmp_seq=5 ttl=59 time=55.527 ms
```

PC5> ping 192.168.1.10

```
PC5> ping 192.168.1.10

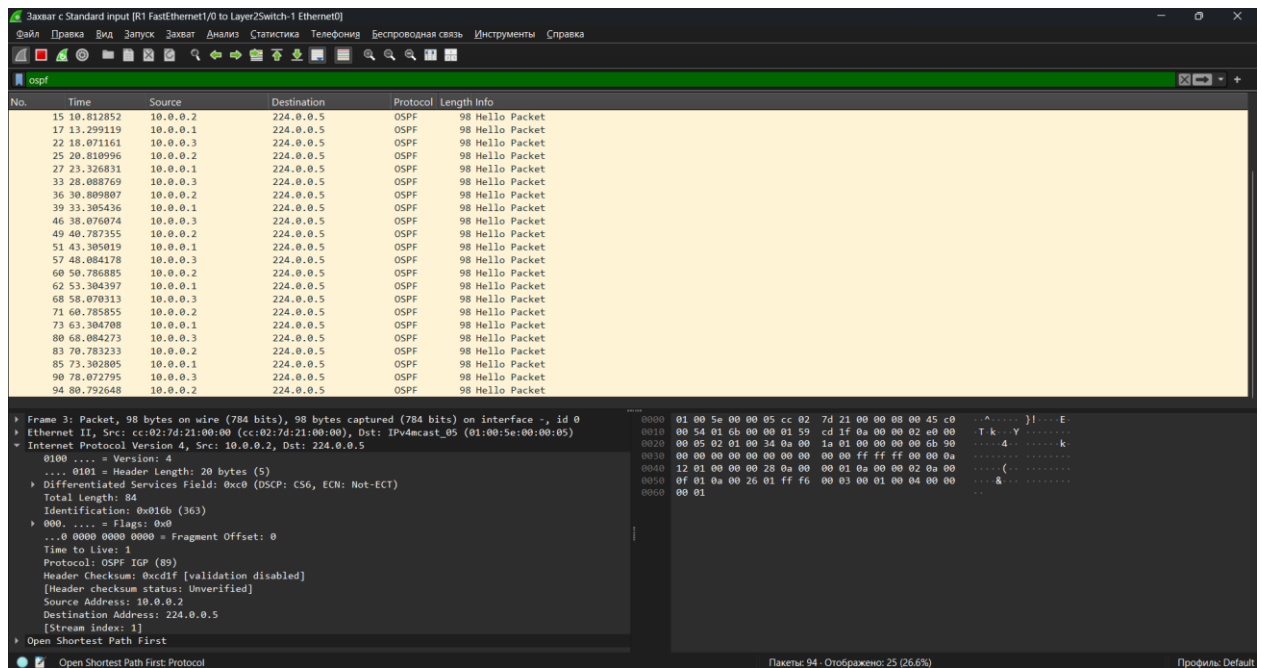
84 bytes from 192.168.1.10 icmp_seq=1 ttl=59 time=69.443 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=59 time=55.781 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=59 time=55.752 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=59 time=55.620 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=59 time=56.184 ms
```

6) Перехватить в wireshark сообщения протоколов RIP v2 и OSPF, идентифицировать их тип и содержание.

The screenshot shows a Wireshark capture of network traffic on the interface Standard input [R4 FastEthernet1/0 to R5 FastEthernet1/0]. The filter applied is `udp.port == 520`. The packet list shows several RIPv2 messages. The selected packet (Frame 5) is a RIPv2 Response from 10.0.45.1 to 224.0.0.9. The packet details pane shows the following structure:

- Frame 5: Packet, 66 bytes on wire (528 bits), 66 bytes captured
- Ethernet II, Src: cc:04:7d:5d:00:10 (cc:04:7d:5d:00:10), Dst: IP
- Internet Protocol Version 4, Src: 10.0.45.1, Dst: 224.0.0.9
- User Datagram Protocol, Src Port: 520, Dst Port: 520
- Routing Information Protocol
 - Command: Response (2)
 - Version: RIPv2 (2)
 - IP Address: 192.168.1.0, Metric: 1
 - Address Family: IP (2)
 - Route Tag: 0
 - IP Address: 192.168.1.0
 - Netmask: 255.255.255.0
 - Next Hop: 0.0.0.0
 - Metric: 1

The packet bytes pane shows the raw data in hexadecimal and ASCII format.



7) Сохранить в отдельные файлы с префиксом `rt_` и именем маршрутизатора таблицы маршрутизации всех маршрутизаторов.

`show ip route`

8) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств.

`show running-config`